

Tables

Table 1: List of Factors for Cross-Sectional Asset Pricing

Factor ID	Description	Reference
Panel A: Credit Risk Factors		
BTP_BUND	Yield differential between the Italian 10-year BTP and the German Bund.	Fausch and Sigo-nius (2018)
CDX_IG	First difference of the CDS spread on the CDX Investment Grade index.	Klaus and Rzep-kowski (2009)
CREDIT_EU	Same construction as CREDIT_US, computed using Bloomberg euro-area corporate bond yield indices for BBB and AAA ratings.	Fama and French (1989)
CREDIT_US	Bond credit risk factor. Yield spread between Moody's BAA and AAA corporate bond indices.	Fama and French (1989)
DEF_EU	Same construction as DEF_US, computed using long-term Euro-pean corporate bond total returns relative to long-term German Bund total returns.	Fama and French (1993)
DEF_US	Bond default risk factor. Total return spread between a long-term U.S. corporate bond portfolio and a long-term U.S. government bond portfolio.	Fama and French (1993)
EBP	Excess Bond Premium. Residual component of aggregate cor-porate bond spreads after netting out expected default risk and bond characteristics via cross-sectional spread regressions; prox-ies time-varying credit risk compensation tied to intermediary constraints. Used with a one-month lag.	Gilchrist and Za-krajšek (2012)
ITRX_MAIN	First difference of the CDS spread on the iTraxx Europe Main index.	Klaus and Rzep-kowski (2009)
MAIN_5/3_FLATTENER	First difference of the natural logarithm of the ratio between the 5-year and 3-year iTraxx Europe Main CDS spreads.	Market indicator
PB_EU_CDS_1Y	Average 1-year CDS spread of European prime brokers.	Klaus and Rzep-kowski (2009)
PB_EU_CDS_5Y	Average 5-year CDS spread of European prime brokers.	Klaus and Rzep-kowski (2009)
PB_US_CDS_1Y	Average 1-year CDS spread of U.S. prime brokers.	Klaus and Rzep-kowski (2009)
PB_US_CDS_5Y	Average 5-year CDS spread of U.S. prime brokers.	Klaus and Rzep-kowski (2009)

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Factor ID	Description	Reference
SNR-MAIN	First difference of the 5-year CDS spread difference between the iTraxx Senior Financials index and the iTraxx Europe Main index.	Market indicator
XOVER/MAIN	First difference of the natural logarithm of the ratio between the 5-year iTraxx Crossover and 5-year iTraxx Europe Main CDS spreads.	Market indicator
Panel B: Liquidity Factors		
Δ FAILS_ PCT_TSY	First difference of the ratio of REPO_FAILS to U.S. Total Debt Outstanding, expressing repo fails as a share of total marketable U.S. Treasury debt.	Fleckenstein et al. (2014)
EURIBOR_OIS	Euribor–OIS spread. Difference between 3-month Euribor and the 3-month overnight index swap rate based on EONIA or €STR.	Nyborg and Östberg (2014)
GC-REPO_ T-BILL	Spread between the 3-month U.S. Treasury general-collateral repo rate and the 3-month U.S. Treasury bill yield.	Bai and Collin-Dufresne (2019)
HKM_IC	Intermediary capital factor. AR(1) innovation in the market-based capital ratio of primary dealers, scaled by the lagged capital ratio.	He et al. (2017)
HPW_NOISE	Treasury noise measure of market-wide illiquidity. Root mean squared deviation between observed U.S. Treasury yields and model-implied yields from a fitted smooth zero-coupon yield curve, aggregated across bonds with maturities between 1 and 10 years.	Hu et al. (2013)
ILLIQ	Change in the Amihud illiquidity measure. First difference of the ratio of absolute daily return to daily dollar volume, capturing shocks to market illiquidity.	Amihud and Mendelson (2015)
LIBOR_OIS	Libor–OIS spread. Difference between 3-month LIBOR and the 3-month overnight index swap rate.	Nyborg and Östberg (2014)
LIBOR_REPO_ SHOCK	Defined as the AR(2) residual of the spread, where the spread is the monthly average of daily 3-month U.S. interbank LIBOR minus the 3-month U.S. Treasury general-collateral repo rate.	Asness et al. (2013)
LIQ_V	Value-weighted return on a 10–1 portfolio formed by sorting stocks on historical liquidity betas.	Pástor and Stambaugh (2003)
LIQNT	Pastor–Stambaugh non-traded liquidity factor. Innovation in aggregate market liquidity constructed from a cross-sectional average of stock-level order-flow-induced return-reversal measures, with predictable components removed.	Pástor and Stambaugh (2003)

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Factor ID	Description	Reference
REPO_FAILS	Repo fails measure. Monthly total notional amount of U.S. Treasury securities fails-to-deliver reported by primary dealers, excluding TIPS, expressed in USD billions.	Fleckenstein et al. (2014)
SILLIQ	Stock illiquidity shock. Innovation in the monthly stock-market illiquidity series, defined as the AR(3) residual of the aggregate Amihud illiquidity measure.	Acharya et al. (2013)
TED_SHOCK_EU	Same construction as TED_SHOCK_US, computed using the monthly average of daily 3-month Euribor minus the 3-month German government bill rate.	Asness et al. (2013)
TED_SHOCK_US	Defined as the AR(2) residual of the TED spread, where the TED spread is the monthly average of daily 3-month U.S. interbank LIBOR minus the 3-month U.S. government bill rate.	Asness et al. (2013)
Panel C: Equity Factors		
BAB_EU	Betting Against Beta (Europe). Same construction as BAB_US, implemented on the European equity universe.	Frazzini and Pedersen (2014)
BAB_US	Betting Against Beta (US). Long low-beta stocks and short high-beta stocks; each leg is rescaled to unit beta, forming a near market-neutral zero-cost factor.	Frazzini and Pedersen (2014)
CMA_EU	Investment factor (Conservative Minus Aggressive). Return spread between European conservative and aggressive investment portfolios from independent size-investment sorts.	Fama and French (2017)
GMOM_EU	Global momentum factor. Return spread between recent winner and loser equity portfolios based on past 12-month returns excluding the most recent month, expressed in euros.	Asness et al. (2013)
GVAL_EU	Global value factor. Return spread between high and low book-to-market equity portfolios formed within major developed equity regions, expressed in euros.	Asness et al. (2013)
HML_EU	Value factor (High Minus Low). Return spread between European high and low book-to-market portfolios from independent size-B/M sorts, approximately size-neutral.	Fama and French (2017)
MKT_EU	Market excess return on the European value-weighted equity market portfolio over the 1-month German government bill. Returns are expressed in euros.	Fama and French (2017)
RMW_EU	Profitability factor (Robust Minus Weak). Return spread between European robust and weak operating profitability portfolios from independent size-profitability sorts.	Fama and French (2017)

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Factor ID	Description	Reference
SMB_EU	Size factor (Small Minus Big). Return spread between diversified European small- and large-cap portfolios from independent size sorts (five-factor construction).	Fama and French (2017)
UMD_EU	Momentum factor (Up Minus Down). Monthly return on a European zero-investment winners-minus-losers momentum portfolio.	Carhart (1997)
Panel D: Volatility Factors		
$\Delta V2X$	Change in V2X. Monthly first difference of the V2X index, the European counterpart of VIX measuring 30-day implied volatility on Euro STOXX 50 options.	Chung et al. (2019)
ΔVIX	Change in VIX. Monthly first difference of the VIX index, a model-free measure of 30-day implied volatility on S&P 500 options.	Chung et al. (2019)
ATM_IV_CDX	3-month at-the-money implied volatility on options written on the CDX Investment Grade index.	Novales and Chamizo (2019)
ATM_IV_ITRX	3-month at-the-money implied volatility on options written on the iTraxx Europe Main index.	Novales and Chamizo (2019)
EP_SVIX_1M	Equity premium proxy from the SVIX methodology. Option-implied risk-neutral variance of the market simple return over the next month, computed by integrating out-of-the-money index option prices across strikes.	Martin (2017)
EP_SVIX_3M	Same construction as EP_SVIX_1M, using options with a 3-month horizon.	Martin (2017)
IV_BUND	3-month at-the-money implied volatility on the 10-year German Bund futures options, based on 100% moneyness.	Cremers et al. (2021)
IV_TSY	3-month at-the-money implied volatility on the 10-year U.S. Treasury futures options, based on 100% moneyness.	Cremers et al. (2021)
MOVE	Option-implied U.S. rates volatility. Yield-curve weighted average of at-the-money implied normal yield volatilities from OTC options on 2-, 5-, 10-, and 30-year constant-maturity interest rate swaps; used lagged by one month.	Bansal and Stivers (2022)
MOVE2	Squared MOVE; used lagged by one month to capture nonlinear exposure to option-implied U.S. interest rate volatility.	Bansal and Stivers (2022)
PTFSBD	Return on the bond trend-following factor, constructed from lookback straddles on government bond futures.	Fung and Hsieh (2001)
PTFSCOM	Return on the commodity trend-following factor, constructed from lookback straddles on commodity futures.	Fung and Hsieh (2001)

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Factor ID	Description	Reference
PTFSFX	Return on the currency trend-following factor, constructed from lookback straddles on major currency futures.	Fung and Hsieh (2001)
PTFSIR	Return on the short-term interest rate trend-following factor, constructed from lookback straddles on 3-month interest rate futures.	Fung and Hsieh (2001)
PTFSSTK	Return on the equity index trend-following factor, constructed from lookback straddles on stock index futures.	Fung and Hsieh (2001)

Panel E: Macro Factors

ΔUF	Change in financial uncertainty. First difference of the financial uncertainty index, constructed as ΔUM but using a panel of financial market variables.	Ludvigson et al. (2021)
ΔUM	Change in macroeconomic uncertainty. First difference of the macro uncertainty index, a model-based measure of expected volatility in the unforecastable component of a large panel of macroeconomic variables.	Ludvigson et al. (2021)
ΔUR	Change in real uncertainty. First difference of the real-activity uncertainty subindex, constructed from real activity variables.	Ludvigson et al. (2021)
5Y5YREAL_YIELD	First difference of the spread between the 5y5y interest rate swap rate and the 5y5y inflation swap rate.	Market indicator
BFCI_EU	Bloomberg Euro Area Financial Conditions Index. Composite Z-score of euro-area money, bond, and equity indicators, standardized to the 1999 to July 1, 2008 pre-crisis period and expressed in standard deviations from pre-crisis conditions.	Osina (2019)
EPU_EU	Economic Policy Uncertainty index for Europe. Same construction as EPU_US, implemented on European newspapers and aggregated across European countries.	Baker et al. (2016)
EPU_US	Economic Policy Uncertainty index for the United States. Newspaper-based index constructed from the normalized frequency of articles that jointly reference the economy, uncertainty, and policy-related terms.	Baker et al. (2016)

Panel F: Interest Rate Factors

$\Delta SLOPE_EU$	Same construction as $\Delta SLOPE_US$, computed using the 10-year German government bond yield minus the 3-month German government bill yield.	Ferson and Harvey (1991)
$\Delta SLOPE_US$	First difference of the U.S. yield-curve slope, defined as the 10-year Treasury yield minus the 3-month Treasury bill yield.	Ferson and Harvey (1991)
EURUSD_3M_IV	First difference of the 3-month at-the-money implied volatility on the EUR/USD exchange rate.	Market indicator

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Factor ID	Description	Reference
SS2Y	Difference between the 2-year EUR fixed swap par rate and the 2-year German Schatz yield.	Collin-Dufresn et al. (2001)
SS5Y	Difference between the 5-year EUR fixed swap par rate and the 5-year German Bobl yield.	Collin-Dufresn et al. (2001)
SS10Y	Difference between the 10-year EUR fixed swap par rate and the 10-year German Bund yield.	Collin-Dufresn et al. (2001)
TERM_EU	Same construction as TERM_US, computed using long-term German Bund total returns in excess of the 1-month German government bill return.	Fama and French (1993)
TERM_US	Bond term structure factor. Excess total return on long-term U.S. government bonds over the 1-month U.S. Treasury bill return.	Fama and French (1993)
YSP_EU	Yield-curve slope factor. Difference between the 5-year and 1-year German government bond yields.	Koijen et al. (2017)
YSP_US	Yield-curve slope factor. Difference between the 5-year and 1-year U.S. Treasury yields.	Koijen et al. (2017)

Note: This table lists all factors used in the cross-sectional asset pricing analysis. Unless otherwise noted, all factor data are obtained from Bloomberg Terminal. Factor IDs correspond to the naming convention used throughout the paper.